Mathematics HL

Internal Assessment

“AtTENTion: A Mathematical Investigation of Conical Tents”

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1. Introduction

In a world where global warming and climate change are moving more in the focus of society, sustainability and environmental awareness is dramatically increased, also in the tourism industry. Individuals want to continue traveling but in a more ecological friendly manner with having less negative environmental impact. Tents are thus becoming a significant ecological alternative, so people are able to adjust their lifestyle.

Last summer I went on a trip with my friend group, but we had to bring a multiplicity of tents, as most are only produced to provide room for a small quantity of people. One of my friends brought a tent in an unusual shape that he had laid out before building it (*Figure 1*). I recognized it to be a circle with a sector cut out as in *Figure 2* and when it was set up it became of a conical shape. However, there was much unused space due to it being very tall and the space in the top could not be used for sleeping capacities. This sparked my interest in how it would be able to achieve the maximum volume for tents like these to maximize the potential of people to fit into because this would simplify travelling ecologically friendly in large groups.

Fig. 1: "Robens Green Cone Tent"

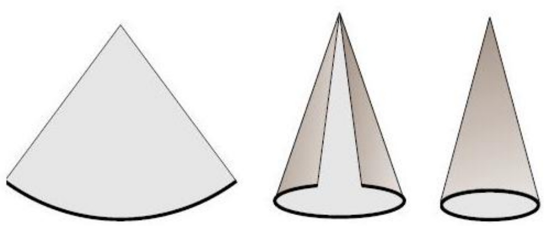


Fig. 2: Circular Cone

The aim of this exploration will therefore be to find out the maximum volume of such. In order to ensure the least waste and largest volume (with respect to the radius) possible, a conical tent can be created by a specific angle being cut out, which is the deciding factor of the potential volume. The calculations within this exploration will derive the formula of the volume of a cone, find the angle of the circle that ensures the maximum volume of a cone, as well as comparing the surface area and volume of cones and triangular prisms. Although it might seem counter-intuitive due to their antagonistic nature, this investigation aims to achieve this through differential calculus and integration. Integration will be needed to derive the formula for the volume of the cone and with differential calculus will be utilized to find the optimum angle for the maximum volume. The mathematics will regard a very specific form of tents (cones) due to a large variation available and in order to obtain a general idea of what is required to maximize their volume. A recommendation for the optimum angle will be the main result of this investigation.

2. Deriving the volume of a cone formula

In order to create a cone out of a circle, a sector must be cut out. The angle of this sector and the respective leftover angle of the circle (θ) will determine the possible volume of the cone. Therefore, the radius of the initial circle (R) will become the slant height of the new cone created.

Cones are objects of rotational symmetry with a circular cross-section and curved sides. A cone, for example, can be made by rotating the graph of a linear function about the x-axis through 360°. The area that is going to be rotated around the x-axis will be half of the triangle cross section of the cone, which is the shaded cross-section of the triangle. When translating this to a graphical representation, the constants "r" (radius of circular cross-section) and "h" (height of cone) are the required constants, as these are needed in the formula for volume (*Figure 4*). The boundaries of the x-axis must therefore correspond to the height of the cone, since we add the areas of all circles within the cone from 0 to h to find the volume. The boundaries of the y-axis must correspond with the radius of it, resulting in the following graphical representation. By taking the equation of the line, f(x) can be found and inserted into the formula of the volume of the solid of revolution. When the line is spun around the x-axis, the cone is formed. The equation of the line spun around the circle is $f(x) = \frac{r}{h}x$. Hence, $f(x)$ is plugged into the formula calculating solid of revolution with the respective boundaries.

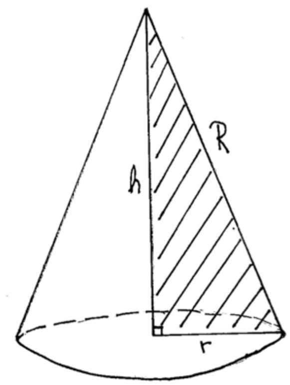


Figure 3

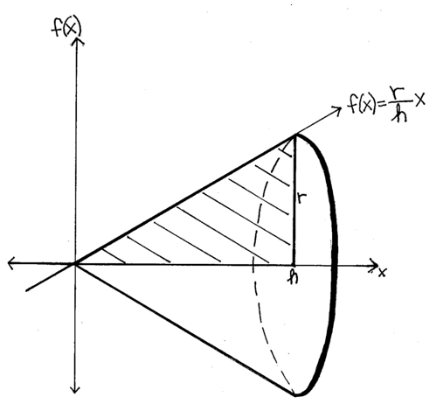


Figure 4: Area of triangle spun around x-axis

$$V = \pi \int_0^h \left(\frac{r}{h}x\right)^2 dx$$

$$= \pi \int_0^h \frac{r^2}{h^2} x^2 dx$$

$$= \frac{\pi r^2}{h^2}\int_0^h x^2 dx$$

$$= \frac{\pi r^2}{h^2}\left|\frac{x^3}{3}\right|\begin{matrix} h \\ 0 \end{matrix}$$

$$= \frac{\pi r^2}{h^2} \times \frac{h^3}{3} - 0 = \frac{\pi r^2 h^3}{h^2 3}$$

$$\boldsymbol{V = \frac{1}{3}\pi r^2 h}$$

3. Calculating the angle to get maximum volume

However, the volume of a cone is treated as a given in any formula booklet or mathematic investigation. I therefore decided to find what the maximum volume is that can be achieved, specifically in relation to the optimum angle of the remaining of the original circle (θ), as is evident in *Figure 3*.

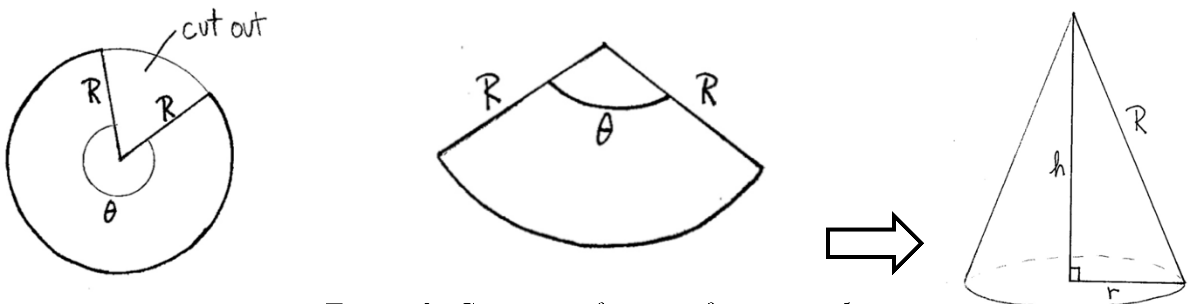


Figure 3: Creation of a cone from a circle

In order to find the maximum volume of the cone, differentiation is necessary, which might seem counter-intuitive, due to having used integration to arrive at the needed formula. The stationary points of graphs are either a maximum or minimum or a point of inflexion. A point $(a, f(a))$ is a possible extremum if $f'(x = a) = 0$. However, the second derivative condition $f''(x = a) < 0$ must also be fulfilled to show we do indeed have a maximum point at $x = a$.

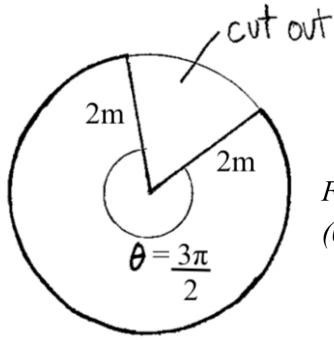
Hence, in order to find the maximum, the first and second derivative must be taken with respect to θ due to which the volume formula must only include this variable. Since the formula for the volume of cones is

$V = \frac{1}{3}\pi r^2 h$ we must replace r^2 and h in terms of θ.

To find *r*, we consider the formula for the circumference of the circle ($C = 2\pi r$) and can see, that $2\pi r = \theta R$, as θ replaces 2π when a sector is cut out of the original circle (**Figure 1**).

When solving for r we thus get: $r = \frac{\theta R}{2\pi}$

This relationship can be exemplified by the following: If we assume the real-life application of tents and constant R = 2 meters and $\theta = \frac{3\pi}{2}$, then the radius of the cone would be calculated by:



$$r = \frac{\frac{3\pi}{2}\, x\, 2}{2\pi} = 1.5\, meters$$

Figure 3: Initial circle ($\theta = \frac{3\pi}{2}$, R=2m)

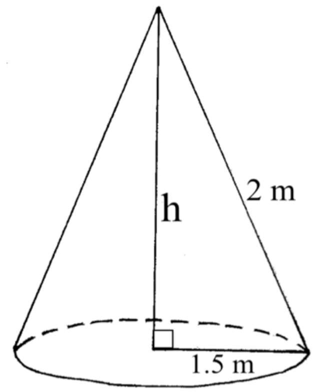


Figure 4: Cone created from $\theta = \frac{3\pi}{2}$, R=2m

Due to the height (h), radius of the base of the cone (r), and the slant length (R) having the following Pythagorean relationship, we can make use of this and solve for h:

$$r^2 + h^2 = R^2$$

$$h = \sqrt{R^2 - r^2}$$

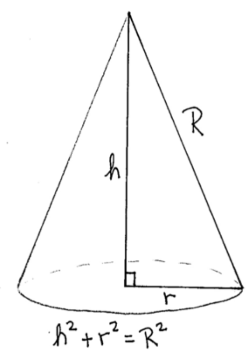


Figure 5: Pythagorean identity of a cone

By knowing R to be a constant and having solved *r* for θ, we can now do the same for h and replace both *r* and *h* in the volume formula

$$h = \sqrt{R^2 - r^2}$$

$$h = \sqrt{R^2 - \left(\frac{\theta R}{2\pi}\right)^2}$$

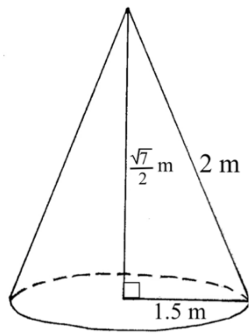
$$V = \frac{1}{3}\pi r^2 h$$

$$V(\theta) = \frac{1}{3}\pi\left(\frac{\theta R}{2\pi}\right)^2 \sqrt{R^2 - \left(\frac{\theta R}{2\pi}\right)^2} = \frac{\pi\theta^2 R^2}{12\pi^2}\sqrt{R^2\left(\frac{4\pi^2 - \theta^2}{4\pi^2}\right)}$$

$$\boldsymbol{V(\theta) = \frac{\theta^2 R^3}{12\pi}\sqrt{\frac{4\pi^2 - \theta^2}{4\pi^2}}}$$

We can apply this to our previous example of $\theta = \frac{3\pi}{2}$:

$$h\left(\frac{3\pi}{2}\right) = \sqrt{2^2 - \left(\frac{\frac{3\pi}{2}\, x\, 2}{2\pi}\right)^2} = \frac{\sqrt{7}}{2} \approx 1.323\, m$$



$$V\left(\frac{3\pi}{2}\right) = \frac{1}{3}\pi\left(\frac{\frac{3\pi}{2}x2}{2\pi}\right)^2\sqrt{2^2 - \left(\frac{\frac{3\pi}{2}x\ 2}{2\pi}\right)^2} \approx\ 3.117\ m^3$$

Figure 6: Cone made from sector with angle $\theta = \frac{3\pi}{2}$

Since we now have the volume of a cone only in terms of one variable (θ) we can differentiate it to find the value of θ that gives the maximum volume.

$$V'(\theta) = \frac{d}{d\theta}\left(\frac{\theta^2 R^3}{12\pi}\sqrt{\frac{4\pi^2-\theta^2}{4\pi^2}}\right)$$

$$= \frac{R^3}{24\pi^2}x\frac{d}{d\theta}\left(\theta^2\sqrt{4\pi^2-\theta^2}\right)$$

$$= \frac{R^3}{24\pi^2}\left[\left(2\theta\sqrt{4\pi^2-\theta^2}\right) + \left(\frac{\theta^2(-2\theta)}{2\sqrt{4\pi^2-\theta^2}}\right)\right]$$

$$= \frac{R^3}{24\pi^2}\left[\left(\frac{2\theta(4\pi^2-\theta^2}{\sqrt{4\pi^2-\theta^2}}\right) - \left(\frac{\theta^3}{\sqrt{4\pi^2-\theta^2}}\right)\right]$$

$$\boldsymbol{V'(\theta) = \frac{R^3}{24\pi^2}\left(\frac{8\pi^2\theta - 3\theta^3}{\sqrt{4\pi^2-\theta^2}}\right)}$$

Now we must solve $\boldsymbol{V'(\theta) = 0}$ to find possible θ values of the stationary points of $V(\theta)$.

$$0 = \frac{R^3}{24\pi^2}\left(\frac{8\pi^2\theta - 3\theta^3}{\sqrt{4\pi^2-\theta^2}}\right)$$

From this we obtain the results $\theta \approx 0\ or \pm 5.13\ radians$

We can reject 0 and -5.13 radians because a volume cannot be negative or equal to zero. Therefore, the optimum angle of the original circle sector is about 5.13 radians. To ensure that this is in fact a maximum, not any other stationary point, we must take the second derivative.

$$\frac{d^2V}{d\theta^2} = \frac{R^3}{24\pi^2}\times\frac{d^2}{d\theta^2}\left(\frac{8\pi^2\theta - 3\theta^3}{\sqrt{4\pi^2-\theta^2}}\right)$$

Using the quotient rule we obtain:

$$\frac{d^2V}{d\theta^2} = \frac{R^3}{24\pi^2}x\left[\frac{(8\pi^2-9\theta^2)\left(\sqrt{4\pi^2-\theta^2}\right) - \left(\frac{1(-2\theta)}{2\sqrt{4\pi^2-\theta^2}}\right)(8\pi^2\theta - 3\theta^2)}{\left(\sqrt{4\pi^2-\theta^2}\right)^2}\right]$$

$$= \frac{R^3}{24\pi^2} x \left[\frac{\left(\frac{32\pi^4 - 36\theta^2\pi^2 + 6\theta^4}{\sqrt{4\pi^2 - \theta^2}} \right)}{4\pi^2 - \theta^2} \right]$$

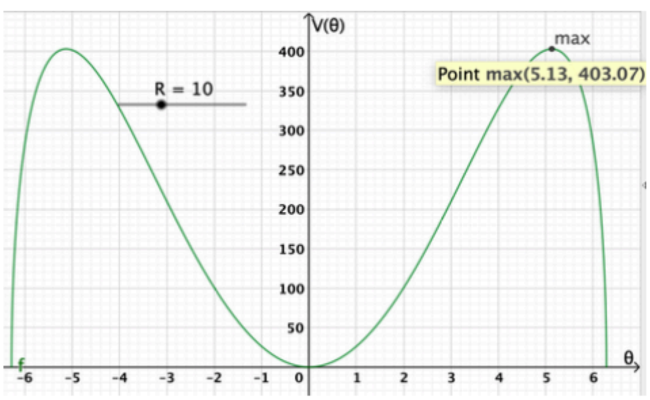
$$\boldsymbol{V''(\theta) = \frac{R^3}{24\pi^2} x \left[\frac{32\pi^4 - 36\theta^2\pi^2 + 6\theta^4}{(4\pi^2 - \theta^2)\sqrt{4\pi^2 - \theta^2}} \right]}$$

According to the second derivative test, the validity of a stationary point is checked by substituting the value obtained (by equating the first derivative to 0) into the second derivative. If $V''(\theta) > 0$, the value obtained represents a minimum. If $V''(\theta) < 0$, the value obtained is a maximum.

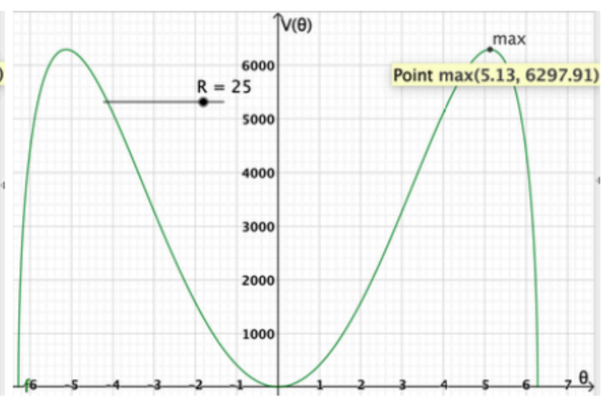
$$V''(5.13) = \frac{R^3}{24\pi^2} x \left[\frac{32\pi^4 - 36(5.13)^2\pi^2 + 6(5.13)^4}{(4\pi^2 - (5.13)^2)\sqrt{4\pi^2 - (5.13)^2}} \right]$$

$$\boldsymbol{V''(5.13) \approx -577.1}$$

This proves that the maximum point for any volume of a cone is obtained when $\theta \approx 5.13$, which is graphically supported. When plotting the graph for $V(\theta)$, the same shape is obtained with the constant R taking any numerical value.



Graph 1: R=10 (created using GeoGebra)



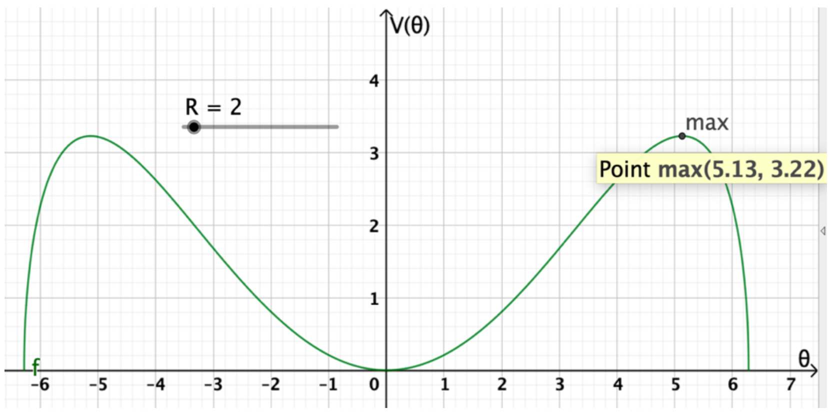
Graph 2: R=25 (created using GeoGebra)

Both maxima for R=10 and R=25 have θ = 5.13 as the x-value for the maximum point. Although the graphs obtained are symmetrical about the y-axis, the maxima with negative values can be rejected since a volume cannot be negative. We must therefore limit the domain to $x \in \mathbb{Z}^{++}$. Thus, the maximum volume of a cone can be obtained when the sector has $\boldsymbol{\theta \approx 5.13}$ **radian**, meaning the removed complementary sector must have an angle of $\theta \approx 1.15$ radian to create a cone with the maximum volume.

The maximum volume obtained therefore is:

$$V(5.13) = \frac{(5.13)^2 R^3}{12\pi}\sqrt{\frac{4\pi^2-(5.13)^2}{4\pi^2}}$$

$$= R^3\left(\frac{(5.13)^2}{12\pi}\sqrt{\frac{4\pi^2-(5.13)^2}{4\pi^2}}\right) \approx 0.4031 \times R^3 \text{ units}^3$$



By looking at *Graph 1* this can be validated, as $V(5.13) \approx 403.1$ if R=10 because $0.4031 \times (10)^3 \approx 403.1$ units^3

To validate the entire process, I examplify this by completing the calculations with $\theta = 5.13$

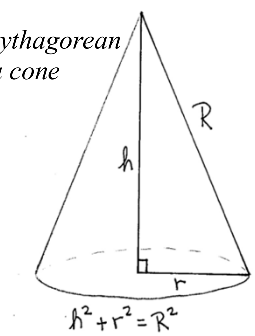
Graph 3: R=2 (created using GeoGebra)

I will use R= 2 m as an example, although it will work with any constant for R.

$$V(5.13) = \frac{(5.13)^2 2^3}{12\pi}\sqrt{\frac{4\pi^2-(5.13)^2}{4\pi^2}}$$

$V(5.13) \approx 3.224$ units^3, which correlates with *Graph 3*.

Figure 7: Pythagorean identity of a cone



4. Alternative method of obtaining optimum angle

During my investigation I also came across another way of obtaining the formula for acquiring the maximum volume of a cone. This, however, calculates the value of θ for the sector to be cut out of the original circle, rather than the angle of the sector that is left over. By having solved for r^2, we can then replace this in the equation of the volume of the cone.

$$R^2 = h^2 + r^2$$

$$r^2 = R^2 - h^2$$

$$V = \frac{\pi}{3}r^2 h = \frac{\pi}{3}(R^2 - h^2)h$$

In order to maximize the volume, we must find the optimum height. To do this, we take the derivative of both sides with respect to h.

$$\frac{dV}{dh} = \frac{\pi}{3} x \frac{d}{dh}(R^2h - h^3)$$

$$\frac{dV}{dh} = \frac{\pi}{3}(R^2 - 3h^2)$$

Setting this equation equal to 0, we find the stationary points, thus the maximum volume.

$$0 = \frac{\pi}{3}(R^2 - 3h^2)$$

$$h^2 = \frac{R^2}{3}$$

$$h = \pm\frac{R}{\sqrt{3}} \qquad \text{(reject negative value, as height can only be positive)}$$

Now that we have found the value for h in terms of the constant R, we must do the same for r^2, so that we can work with only the constant in the final equation for the volume of the cone.

$$r^2 = R^2 - h^2$$

$$r^2 = R^2 - \frac{R^2}{3}$$

$$r^2 = \frac{2}{3}R^2$$

Now both variables in the equation can be replaced: $V = \frac{1}{3}\pi\left(\frac{2}{3}R^2\right)\left(\frac{1}{\sqrt{3}}R\right)$

We can now come up with the final equation that can determine the maximum volume for any constant R (radius of the initial circle):

$$V = R^3\left(\frac{2\pi}{9\sqrt{3}}\right)$$

However, to come up with an answer to the question regarding the optimal section to cut out we must use the formula of the circumference of the base of the cone. Let θ be the section to be cut.

$$2\pi R - \theta (= 2\pi r) = 2\pi\sqrt{\frac{2}{3}}R$$

$$2\pi R - \theta = 2\pi\sqrt{\frac{2}{3}}R$$

$$\theta = 2\pi R\left(1 - \sqrt{\frac{2}{3}}\right)$$

$$\frac{\theta}{R} = 2\pi\left(1 - \sqrt{\frac{2}{3}}\right) \approx 1.15\ radians$$

Maximum volume can thus be achieved when the angle of the sector cut out equals 1.15 radians.

To test this, we use the previous example of R=10. This would result in the following equation:

$$\theta = 2\pi(10)\left(1 - \sqrt{\frac{2}{3}}\right) \approx 11.5$$

$$\frac{\theta}{R} = \frac{11.5}{10} \approx 1.15$$

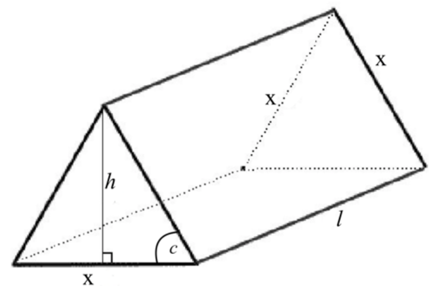


Figure 8: Triangular Prism

5. Applicability to more common tent shape

The sphere is known to be the more effective 3D shape, in that it entails the most volume in the least surface area (lowest SA/V ratio). However, this is not a fitting shape in the context of tents. The more common tent shape is a triangular prism, which can now be compared to the SA/V ratio to cones.

It is vital that we also find out how much surface area this means for a tent of this size in order to clarify the requirements for future productions. In the context of the tents, it is vital to consider the surface area that is established by the maximum volume and to include the base of the "cone" in the calculations. For this we make use of the surface area formula.

Curved surface area of a cone = $\pi r l$, in which l means the slant length (R in our case) resulting in the equation: **Surface area of cone (SA) =** $\boldsymbol{\pi r R}$. To this we must then add the ***Area of the base (AB)*** **=** $\boldsymbol{\pi r^2}$. Continuing with the example of R= 2m, we thus calculate the following:

SA + AB = Total surface area (TSA)

$$\pi r R + \pi r^2 = TSA$$

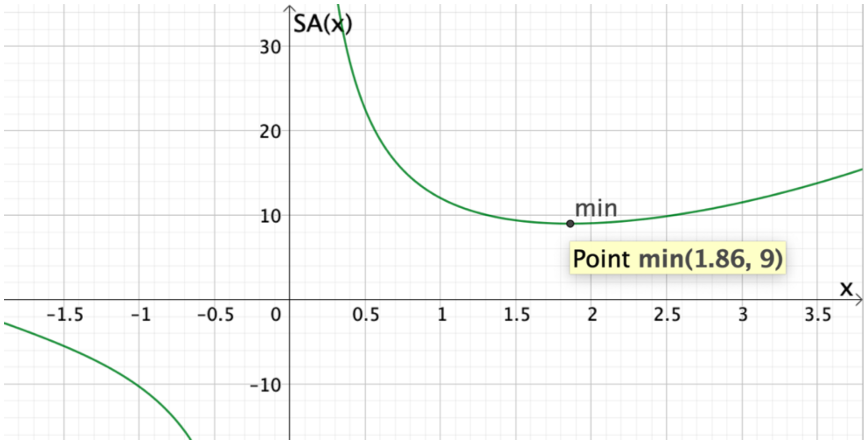
$$(\pi\ x\ 1.5\ x\ 2) + (\pi\ x\ (1.5)^2) \approx 16.49\ m^2$$

However, the more common shapes for tents are A-Frame tents, seen in *Figure 8*, which equals the geometric shape of a triangular prism. This raises the question how much material you need to cover the surface area if you want to obtain the equivalent volume, especially compared to that compared to the SA of the cone with the same volume. To be able to calculate we must use the simplest form of triangles as the base: equilateral triangles. this, the formula for the volume is

$$V = l\left(\frac{1}{2}x^2 sinc\right) = l\left(\frac{1}{2}x^2 sin\frac{\pi}{3}\right)$$

$$V = l\left(\frac{\sqrt{3}}{2}x^2\right)$$

To compare the Volume to SA ratio between cones and triangular prisms, we can exemplify this with the example of R=2m and $\theta = 5.13\, radians$, which resulted in Volume=3.224 me^3 and SA=16.49 m^2. By equating this volume with the formula for the triangular prism we obtain:



Graph 4: Minimum point of SA(x) of Triangular Prism with V= 3.224 units^3

$$3.224 = l\left(\frac{\sqrt{3}}{2}x^2\right)$$

When solving for l, we get $l = \frac{6.448}{\sqrt{3}x^2}$

To find the respective surface area we must use substitution and plug this value for l into the formula of the surface area to ultimately be able to find the minimum amount of material needed.

$$SA = 2\left(\frac{\sqrt{3}}{4}x^2\right) + 3lx$$

$$SA = 2\left(\frac{\sqrt{3}}{4}x^2\right) + 3\left(\frac{6.448}{\sqrt{3}x^2}\right)x = \frac{\sqrt{3}}{2}x^2 + \frac{19.34}{\sqrt{3}x}$$

When graphing this (or utilizing the first and second derivative test), we find that the **minimum** occurs at **(1.86, 9)**, visible in *Graph 3*.

We can also find that the length of the tent is 1.076 meter by using the x-value of the minimum point. The minimum value suggests that the minimum surface area obtained by a triangular prism with a volume of 3.224 meter^3 is 9 meter^2. Moreover, this shows that a triangular prism has less surface area than a triangular prism (regular shape of tents) with this specific volume, as the cone has a SA of 243.84 units^2, making cones a more effective shape when considering the least amount of material to be needed in the production of tents for example.

6. Conclusion

The results of this investigation show, that there are multiple ways of obtaining the optimum remaining angle of the initial circle to achieve the maximum volume of a cone. This entails the process of finding the volume formula in terms of θ and the constant of the slant length/initial radius *R*. Using definite integration the required volume formula can be acquired and by the means of differential calculus, we are thus able to find the optimum value of θ that allows the maximum volume of cones to be obtained with any other constants given. This value for θ proved to be 5.13 radians, although an alternative method also allows the obtaining of the inverse of this value, meaning the angle of the sector that subtracted from the circle. In this case, the derivative of the volume formula is taken with respect to the height, and the variable r is replaced by utilizing the Pythagorean relationship. This has the aim of comparing the circumference of the initial and the altered circle, so that the value of θ can be obtained, which is proven to be 1.15 radians. Since 5.13+1.15 = 6.28 (radians within a full circle), this proves the validity of both methodologies.

Overall, this investigation was of great aid to my understanding of differential and integral calculus, as my methodology demonstrated that they should not be recognized as opposing tools, but rather as complementary. While integration is needed for the creation of the required formula, differentiation allows the finding of stationary points of the respective function.

There is great difference in the SA to Volume ratio of a triangular prism and a cone, as shown in my findings. This demonstrated the geometric significance, as the general potential of 3D shapes can be compared based on this ratio.

Some limitations underlie the nature of this investigation, especially considering I worked with a very simplified shape of a tent: cones: Since tents usually have a rectangular shape, the SA to volume ratio is largely different as evident when comparing the surface area when both shapes have the volume 403 units^3: while the cone has a SA of 243.84 untis^2, the triangular prism has 357.19 units^2. Considering the vast range of available tents, the investigation of cones has very limited applicability, although it does give insight into the potential of cones.

Moreover, the maximum volume does not necessarily equate to the maximum practicability (space for people within the tent). This suggests that there should be more emphasis on increasing the surface area of the base to maximize amount of sleeping space for people, which is not in accordance to maximizing the volume. Although the required angle was found successfully, this might be limited by not applying to all shapes, as the shape of cones is very simplified. Nonetheless, this could be of use by demonstrating the unused potential of tents regarding the saving of material cost. Since the same volume can be achieved with less surface area, the average cost of production is reduced, making tents and sustainable travelling a more appealing alternative.

Another limit of this investigation is that mathematics alone are not able to convince individuals of altering their lifestyle, but rather requires a social shift. No matter the maximum potential calculated, people will remain with their current methods unless they are convinced of a necessary change. However, these calculations can provide the basis for improved products and marketing of the respective businesses.

As an extension to this exploration, one could investigate the potential of different geometric 3D shapes in regard to their SA to volume ratio. This could provide the basis of improved tent shapes and the enhanced effectiveness of the use of space. A comparison of varying shapes can be the grounds to further improvements of eco-tourism and travelling through which the real-life applicability and general relevance can be maximized.

Works Cited:

Fig. 1: “Robens Green Cone Tent.” *Robens Green Cone Tent | Campz.at*, Campz, www.campz.at/robens-green-cone-tent-913386.html.

Fig. 2: “Circular Cone.” *Solid Figures*, Freeganita, www.freeganita.com/en/geo/6_15.htm.

Harnet, Josip, et al. *Mathematics Higher Level*. Oxford University Press, 2012.

Martin, David. *Mathematics for the International Student: Mathematics HL (Core)*. Haese Mathematics, 2012.

“Maximizing the Volume of a Cone.” *Maximizing the Volume of a Cone - Math Central*, University of Regina, 2009, mathcentral.uregina.ca/QQ/database/QQ.09.06/s/christina1.html.

Woo, Eddie. *Conical Volume (1 of 2: Derivation of the Volume of a Cone through Solids of Revolution)*. YouTube, 24 Jan. 2016, www.youtube.com/watch?v=bSTdCgbxc6k.